ZEYU HUANG

zeroyuhuang.github.io/ Google Scholar Page zeroy.huang@gmail.com

Education

School of Informatics, The University of Edinburgh Edinburgh, UK
PhD in NLP; **2026 Apple PhD Scholar in AI/ML** June 2027 expected

Beihang University Beijing, China
Master of Electronic Information; GPA: 3.9/4.0; Ranking: 2/86 June 2023

Beihang University Beijing, China
Bachelor of Information and Computing Science; GPA: 3.7/4.0; Ranking: 6/87 June 2020

Experience

AliStar Research Intern, Qwen Pretraining Team Jul. 2026 - Present
Advisors: Dr. Bo Zheng & Zihan Qiu Beijing

TopSeed Research Intern, ByteDance Seed Speech Team Mar. 2026 - Jul. 2026
Advisors: Dr. Wenhao Zhu & Shanbo Cheng Shanghai

Student Researcher, Google DeepMind Continual Learning Team Aug. 2025 - Feb. 2026
Advisors: Dr. Marc'Aurelio Ranzato & Dr. Adhiguna Kuncoro London

Research Intern at INF Technology Mar. 2024 - Aug. 2024
Advisor: Zili Wang Remote

Research Intern at BAAI Sep. 2022 - Jun. 2023
Advisor: Dr. Jie Fu Beijing, China

Research Intern at WeChat AI Nov. 2021 - May. 2022
Advisor: Dr. Yikang Shen Beijing, China

Research Interests

My research focuses on how large language models learn, including (1) learning paradigms: how models learn from corrections, demonstrations, rewards, and context; (2) optimization objectives: how training objectives shape the behavioral patterns models develop; and (3) model architectures: how modularity, routing in MoE models, and components such as attention and residual connections affect training dynamics.

Publications

* denotes equal contribution.

Zihan Qiu*, **Zeyu Huang***, Kaiyue Wen*, Peng Jin*, Bo Zheng*,..., Ivan Titov, Dayiheng Liu, Jingren Zhou, Junyang Lin. A Unified View of Attention and Residual Sinks: Outlier-Driven Rescaling is Essential for Transformer Training - **COLM 2026 Oral (24 / 2939 submissions)**.

Zeyu Huang, Tianhao Cheng, Zihan Qiu, Zili Wang, Yinghui Xu, Edoardo M Ponti, Ivan Titov. Blending Supervised and Reinforcement Fine-Tuning with Prefix Sampling - **ICML 2026**

Yijun Yang*, **Zeyu Huang***, Wenhao Zhu, Zihan Qiu, Fei Yuan, Jeff Z Pan, Ivan Titov. A Controllable Examination for Long-Context Language Models - **NeurIPS 2025 Spotlight**

Zihan Qiu*, Zekun Wang*, Bo Zheng*, **Zeyu Huang***, Kaiyue Wen, Songlin Yang, ... & Junyang Lin. Gated Attention for Large Language Models: Non-linearity, Sparsity, and Attention-Sink-Free. **NeurIPS 2025 Oral & Best Paper Award**

Zihan Qiu*, **Zeyu Huang***, Bo Zheng, Kaiyue Wen, Zekun Wang, ... & Junyang Lin. Demons in the Detail: On Implementing Load Balancing Loss for Training Specialized Mixture-of-Expert Models - **ACL 2025**

Zeyu Huang, Zihan Qiu, Zili Wang, Edoardo M Ponti, Ivan Titov. Post-hoc Reward Calibration: A Case Study on Length Bias - **ICLR 2025**

Zihan Qiu*, **Zeyu Huang***, Shuang Cheng, Yizhi Zhou, Zili Wang, Ivan Titov, Jie Fu. Layerwise Recurrent Router for Mixture-of-Experts - **ICLR 2025**

Ka Man Lo*, **Zeyu Huang***, Zihan Qiu*, Zili Wang, Jie Fu. A Closer Look into Mixture-of-Experts in Large Language Models. **NAACL 2025 Findings**

Wenyu Du, Tongxu Luo, Zihan Qiu, **Zeyu Huang**, Yikang Shen, Reynold Cheng, Yike Guo, Jie Fu. Stacking Your Transformers: A Closer Look at Model Growth for Efficient LLM Pre-Training - **NeurIPS 2024 Spotlight**.

Wenyu Du, Shuang Cheng, Tongxu Luo, Zihan Qiu, **Zeyu Huang**, Ka Chun Cheung, Reynold Cheng, Jie Fu. Unlocking Continual Learning Abilities in Language Models - **EMNLP 2024 Findings**

Zihan Qiu*, **Zeyu Huang***, Jie Fu. Unlocking Emergent Modularity in Large Language Models. **NAACL 2024**. **Outstanding Paper Award**

Zihan Qiu*, **Zeyu Huang***, Youcheng Huang*, Jie Fu. Empirical Study on Updating Key-Value Memories in Transformer Feed-forward Layers. Tiny Papers @ **ICLR 2024**

Zeyu Huang, Yikang Shen, Xiaofeng Zhang, Jie Zhou, Wenge Rong, Zhang Xiong. Transformer-Patcher: One Mistake Worth One Neuron. **ICLR 2023.**

Zeyu Huang, Wenge Rong, Xiaofeng Zhang, Yuanxin Ouyang, Chenghua Lin, Zhang Xiong. Token Relation Aware Chinese Named Entity Recognition. **ACM TALLIP**

Zeyu Huang, Xiaofeng Zhang, Jun Bai, Wenge Rong, Yuanxin Ouyang, Zhang Xiong. Solving Math Word Problems Following Logically Consistent Template. **IJCNN 2023**

Xiaofeng Zhang*, Yikang Shen*, **Zeyu Huang**, Jie Zhou, Wenge Rong, Zhang Xiong. Mixture of Attention Heads: Selecting Attention Heads Per Token. **EMNLP 2022**

Pre-prints

Zeyu Huang, Adhiguna Kuncoro, Qixuan Feng, Jiajun Shen, Lucio Dery, Arthur Szlam, Marc'Aurelio Ranzato. Context Training with Active Information Seeking

Tianhao Cheng*, **Zeyu Huang***, Zihan Qiu, Yu Cheng, Edoardo Ponti, Yinghui Xu, Ivan Titov, Zenglin Xu. The Cancellation Hypothesis in Critic-Free RL: From Outcome Rewards to Token Credits

Honors and Awards

- **2026 Apple PhD Scholar in AI/ML** (20 worldwide)
- **NeurIPS 2025 Best Paper Award** (4/21575 submissions)
- **NAACL 2024 Outstanding Paper Award** (6/2604 submissions)
- EPSRC DTA Scholarship (international flexibility) - 2024-2027
- National Scholarship for Graduate Students Award - 2021-2022 - (3/291)
- Outstanding Graduate Student of Beihang University - 2020-2021

Extra-curricular Activities

International Exchange — Casablanca, Morocco
Ecole Centrale Casablanca — Nov 2019, 15 days

International Exchange — Paris, France
CentraleSupélec, Saclay — Feb 2018, 8 days

Student works — Beijing, China
President of the Graduate Student Union of ECPK — Dec. 2020 - Dec. 2021

Orange Lab & Ecole Centrale de Pekin — Beijing, China
Oct 2020 - May 2021 — Group Leader of the Project

- **Federated Recommendation** - A recommendation system trained in Federated Learning manner.
- This project is co-supervised by Orange Lab of Beijing and Ecole Centrale de Pekin.
- The product of this project includes a survey report and deployment of a Federated news recommender

Skills Summary

- **Programming**: Python, C, Shell, JAVA, Latex, MATLAB
- **Languages**: Chinese(Native), English (IELTS 7.5), French (B2)
- **ML/AI Frameworks**: PyTorch, JAX, TensorFlow, PyTorch Lightning, Fairseq, Hugging Face, Flax

黄泽宇

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教育背景

爱丁堡大学信息学院，博士（自然语言处理） 2027 年 6 月预计毕业
2026 Apple PhD Scholar in AI/ML 英国爱丁堡

北京航空航天大学，电子信息硕士 2023 年 6 月
GPA 3.9/4.0；专业排名 2/86 中国北京

北京航空航天大学，信息与计算科学学士 2020 年 6 月
GPA 3.7/4.0；专业排名 6/87 中国北京

研究方向

我主要关注大语言模型的学习机制，包括：（1）学习范式：模型如何从纠错、示范、奖励和上下文中学习；（2）优化目标：训练目标如何塑造、影响模型行为模式；（3）模型架构：模块化、MoE 中的路由与注意力、残差等模块如何影响模型的训练过程。

研究与实习经历

阿里星研究实习生，Qwen 预训练团队 2026 年 7 月–至今
指导人：Bo Zheng、Zihan Qiu 中国北京

字节跳动 Seed 语音团队，TopSeed 研究实习生 2026 年 3 月–7 月
指导人：Wenhao Zhu、Shanbo Cheng 中国上海

Google DeepMind 持续学习团队，学生研究员 2025 年 8 月–2026 年 2 月
指导人：Marc'Aurelio Ranzato、Adhiguna Kuncoro 英国伦敦

INF Technology，研究实习生 2024 年 3 月–8 月
指导人：Zili Wang 远程

北京智源人工智能研究院（BAAI），研究实习生 2022 年 9 月–2023 年 6 月
指导人：Jie Fu 中国北京

腾讯微信 AI，研究实习生 2021 年 11 月–2022 年 5 月
指导人：Yikang Shen 中国北京

论文发表

* 表示共同第一作者。论文标题保留正式英文名称。

Zihan Qiu*, **Zeyu Huang***, Kaiyue Wen*, Peng Jin*, Bo Zheng*, et al.. A Unified View of Attention and Residual Sinks: Outlier-Driven Rescaling is Essential for Transformer Training. **COLM 2026 Oral (24 / 2939 submissions)**

Zeyu Huang, Tianhao Cheng, Zihan Qiu, Zili Wang, Yinghui Xu, Edoardo M. Ponti, Ivan Titov. Blending Supervised and Reinforcement Fine-Tuning with Prefix Sampling. **ICML 2026**

Yijun Yang*, **Zeyu Huang***, Wenhao Zhu, Zihan Qiu, Fei Yuan, Jeff Z. Pan, Ivan Titov. A Controllable Examination for Long-Context Language Models. **NeurIPS 2025 Spotlight**

Zihan Qiu*, Zekun Wang*, Bo Zheng*, **Zeyu Huang***, Kaiyue Wen, et al.. Gated Attention for Large Language Models: Non-linearity, Sparsity, and Attention-Sink-Free. **NeurIPS 2025 Oral & Best Paper Award**

Zihan Qiu*, **Zeyu Huang***, Bo Zheng, Kaiyue Wen, Zekun Wang, et al.. Demons in the Detail: On Implementing Load Balancing Loss for Training Specialized Mixture-of-Expert Models. **ACL 2025**

Zeyu Huang, Zihan Qiu, Zili Wang, Edoardo M. Ponti, Ivan Titov. Post-hoc Reward Calibration: A Case Study on Length Bias. **ICLR 2025**

Zihan Qiu*, **Zeyu Huang***, Shuang Cheng, Yizhi Zhou, Zili Wang, Ivan Titov, Jie Fu. Layerwise Recurrent Router for Mixture-of-Experts. **ICLR 2025**

Ka Man Lo*, **Zeyu Huang***, Zihan Qiu*, Zili Wang, Jie Fu. A Closer Look into Mixture-of-Experts in Large Language Models. **NAACL 2025 Findings**

Wenyu Du, Tongxu Luo, Zihan Qiu, **Zeyu Huang**, Yikang Shen, et al.. Stacking Your Transformers: A Closer Look at Model Growth for Efficient LLM Pre-Training. **NeurIPS 2024 Spotlight**

Wenyu Du, Shuang Cheng, Tongxu Luo, Zihan Qiu, **Zeyu Huang**, et al.. Unlocking Continual Learning Abilities in Language Models. **EMNLP 2024 Findings**

Zihan Qiu*, **Zeyu Huang***, Jie Fu. Unlocking Emergent Modularity in Large Language Models. **NAACL 2024 Outstanding Paper Award**

Zihan Qiu*, **Zeyu Huang***, Youcheng Huang*, Jie Fu. Empirical Study on Updating Key-Value Memories in Transformer Feed-forward Layers. **ICLR 2024 Tiny Papers**

Zeyu Huang, Yikang Shen, Xiaofeng Zhang, Jie Zhou, Wenge Rong, Zhang Xiong. Transformer-Patcher: One Mistake Worth One Neuron. **ICLR 2023**

Zeyu Huang, Wenge Rong, Xiaofeng Zhang, Yuanxin Ouyang, Chenghua Lin, Zhang Xiong. Token Relation Aware Chinese Named Entity Recognition. **ACM TALLIP**

Zeyu Huang, Xiaofeng Zhang, Jun Bai, Wenge Rong, Yuanxin Ouyang, Zhang Xiong. Solving Math Word Problems Following Logically Consistent Template. **IJCNN 2023**

Xiaofeng Zhang*, Yikang Shen*, **Zeyu Huang**, Jie Zhou, Wenge Rong, Zhang Xiong. Mixture of Attention Heads: Selecting Attention Heads Per Token. **EMNLP 2022**

预印本

Zeyu Huang, Adhiguna Kuncoro, Qixuan Feng, Jiajun Shen, Lucio Dery, Arthur Szlam, Marc'Aurelio Ranzato. Context Training with Active Information Seeking. **arXiv 2026**

Tianhao Cheng*, **Zeyu Huang***, Zihan Qiu, Yu Cheng, Edoardo Ponti, Yinghui Xu, Ivan Titov, Zenglin Xu. The Cancellation Hypothesis in Critic-Free RL: From Outcome Rewards to Token Credits. **arXiv 2026**

荣誉与奖项

- **2026 Apple PhD Scholar in AI/ML**（全球 20 人）
- **NeurIPS 2025 最佳论文奖**（21,575 篇投稿中获奖论文 4 篇）
- **NAACL 2024 杰出论文奖**（2,604 篇投稿中获奖论文 6 篇）
- EPSRC DTA 奖学金（2024–2027）
- 研究生国家奖学金（2021–2022；年级排名 3/291）
- 北京航空航天大学优秀研究生（2020–2021）

其他经历与技能

学生工作：北京航空航天大学中法工程师学院研究生会主席（2020 年 12 月–2021 年 12 月）；Orange Lab 与中法工程师学院联合项目负责人（2020 年 10 月–2021 年 5 月）。

国际交流：摩洛哥卡萨布兰卡中央理工学院（2019 年 11 月）；法国 CentraleSupélec（2018 年 2 月）。

编程与框架：Python、C、Shell、Java、LaTeX、MATLAB；PyTorch、JAX、TensorFlow、Hugging Face、Flax 等。

语言：中文（母语）、英语（雅思 7.5）、法语（B2）。